

CLASS 11TH CHEMISTRY (THEORY)

ANNUAL BOARD PAPER

Series: 11thARNKD(W/Z) JKLUT-25-1209-C Time: 3.0 Hours

Max. Marks: 70 Marks

Note: Attempt any 60 Marks out of 70 Marks. | Use of log tables is permitted; scientific calculators are not allowed.

SECTION A: MULTIPLE CHOICE QUESTIONS

1 Mark each

Q1(i). A compound with empirical formula C_3H_4N has molecular mass 108 amu. The molecular formula is:

- (A) C_3H_4N (B) $C_6H_8N_2$ (C) $C_9H_{12}N_3$ (D) $C_{12}H_{16}N_4$

Q1(ii). The number of radial nodes in 3s atomic orbital are:

- (A) 2 (B) 0 (C) 1 (D) 3

Q1(iii). Which one of the following metal oxides is most acidic / amphoteric in nature?

- (A) Al_2O_3 (B) Na_2O (C) MgO (D) CaO

Q1(iv). The hybridization of the central iodine atom in IF_7 molecule is:

- (A) sp (B) sp^3d (C) sp^3d^2 (D) sp^3d^3

Q1(v). Which of the following thermodynamic variables is an intensive property?

- (A) Volume (B) Temperature (C) Mass (D) Internal energy

Q1(vi). The hydronium ion concentration $[H_3O^+]$ of a solution is 2×10^{-5} M. Its pH value is:

- (A) 5 (B) 7 (C) 4.7 (D) 3

Q1(vii). Which of the following chemical species is the strongest conjugate base?

- (A) Cl^- (B) SO_4^{2-} (C) CH_3COO^- (D) NO_3^-

Q1(viii). In the redox reaction: $SO_2 + 2HNO_3 \rightarrow H_2SO_4 + 2NO_2$, which of the following is correct?

- (A) HNO_3 is a reducing agent (B) HNO_3 is an oxidising agent
(C) SO_2 is an oxidising agent (D) S atom in SO_2 is reduced

Q1(ix). The oxidation state of carbon atom in carbonate ion (CO_3^{2-}) is:

- (A) -2 (B) +2 (C) -4 (D) +4

Q1(x). Which of the following cyclic hydrocarbon species is aromatic according to Huckel's rule?

- (A) Cyclopropene (B) Cyclopropenyl anion (C) Cyclopropenyl cation (D) Cyclobutadiene

SECTION B: VERY SHORT ANSWER TYPE QUESTIONS

2 Marks each

Q2(i). State any two fundamental limitations / anomalies of Mendeleev's Periodic Table.

Q2(ii). Define solubility product (K_{sp}). Write down the solubility product expressions for:

- (a) $BaCO_3(s) \rightleftharpoons Ba^{2+}(aq) + CO_3^{2-}(aq)$
(b) $Ag_2CrO_4(s) \rightleftharpoons 2Ag^+(aq) + CrO_4^{2-}(aq)$

Q2(iii). Give two balanced chemical equations illustrating redox reactions.

Q2(iv). What is meant by qualitative chemical analysis? Name one standard method used for the estimation of Nitrogen in an organic compound.

Q2(v). What are Nucleophiles? Give two representative examples of nucleophiles in organic reactions.

Q2(vi). What are Carbanions? Give one illustrative example and discuss its hybridization.

Q2(vii) Why are terminal alkynes significantly more acidic than alkenes and alkanes? Explain on the basis of orbital electronegativity.

Q2(viii) Outline the stepwise electrophilic reaction mechanism for the Sulphonation of benzene.

Q2(ix). State Markovnikov's addition rule. Illustrate with the addition of HBr to propene.

SECTION C: SHORT ANSWER TYPE QUESTIONS

3 Marks each

Q3(i). How many grams of NH_3 are produced when 6 grams of H_2 gas reacts with excess of N_2 gas according to Haber's reaction: $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$?

Q3(ii). State and explain the Law of Multiple Proportions with illustrative chemical examples.

Q3(iii). Derive de-Broglie's wave equation ($\lambda = h / mv$) for the dual nature of matter. State its physical significance.

Q3(iv). Define first ionization enthalpy. Explain why ionization enthalpy decreases progressively down an elemental group.

Q3(v). What is VSEPR theory? Explain the bent / V-shaped geometry and bond angle of H_2O on the basis of electron pair repulsions.

Q3(vi). Determine the hybridization of the central atom in each of the following molecules:
(a) XeO_2F_2 (b) SF_6 (c) ICl_3 (d) PCl_5 (e) BeCl_2 (f) CO_2

Q3(vii) What is a thermodynamic State Function? Explain why internal energy (U) is classified as a state function.

Q3(viii) Explain the various physical and chemical factors (concentration, temperature, pressure) that affect the equilibrium state according to Le-Chatelier's principle.

Q3(ix). What are conformational isomers (conformers)? Draw Sawhorse and Newman projections for Eclipsed and Staggered conformations of ethane.

SECTION D: LONG ANSWER TYPE QUESTIONS

5 Marks each

Q4(i). Explain Rutherford's alpha-particle gold-foil scattering experiment. What were the key conclusions and major limitations of Rutherford's nuclear model of atom?

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What is an Atomic Orbital? Describe the shapes, nodal planes, and boundary surface diagrams of s, p, and d orbitals.

Q4(ii). Define molar heat capacity at constant pressure (C_p) and constant volume (C_v). Derive Mayer's thermodynamic relation: $C_p - C_v = R$.

--- OR ---

What is standard enthalpy of formation ($\Delta_f H$)? State Hess's Law of Constant Heat Summation and explain its utility with a Born-Haber cycle example.

Q4(iii). What is hyperconjugation (Baker-Nathan effect)? Explain the relative stability order of primary, secondary, and tertiary carbocations on the basis of hyperconjugation.

--- OR ---

Write the systematic IUPAC names of the following organic compounds:

- (a) $\text{CH}_3\text{-CH}(\text{CH}_3)\text{-CH}(\text{OH})\text{-C}(\text{CH}_3)_2\text{-CH}_3$
- (b) $\text{CH}_3\text{-CH}(\text{OH})\text{-CH}(\text{OH})\text{-CH}_3$
- (c) 2-methylphenol (o-cresol)
- (d) $\text{CH}_3\text{-O-CH}_2\text{-CH}(\text{CH}_3)\text{-CH}_3$
- (e) $\text{HO-CH}_2\text{-CH}_2\text{-CH}_2\text{-COOH}$